JEE Mathematics: Quick Review with Grading and Practice Map - Solution of Triangle (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (23 questions; a question that appears in two of these is counted once): High = 2 or more questions, Medium = 1 to 1 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction about Triangle

Question 01

What is the relation between sides of a triangle?

Model Answer

Sum of any two sides > third side, i.e. $|b-c| < a < b+c$ (same for every side); also $A+B+C=\pi$.

Ex: $2, 3, 6$ cannot be sides, since $2+3<6$.

Grade - Medium

Practice map, main concept: Sheet: C-I 21 | DPP: 2(7)

Also used in: Sheet: J-I 6, 21; J-A 9

Question 02

What do you mean by solving a triangle?

Model Answer

Finding the unknown sides and angles from three given independent elements (at least one of them a side).

Ex: given a, b, C: find c by the cosine rule, then A, B by the sine rule.

Grade - Low

Practice map, main concept: none

Also used in: none

Basic Laws : Sine Law

Question 03

What is sine law?

Model Answer

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R$$

Sides are proportional to the sines of the opposite angles (R = circumradius).

Ex: $a=5,\ A=30^\circ \Rightarrow 2R=\dfrac{5}{1/2}=10.$

Grade - High

Practice map, main concept: Sheet: J-I 1, 7, 15, 17; J-II 33; J-M 8; C-I 8 | DPP: 2(8)

Also used in: Sheet: J-M 2, 6; J-A 11

Question 04

How to express a in terms of $\sin A$?

Model Answer

$$a = 2R\sin A = k\sin A, \quad b = 2R\sin B, \quad c = 2R\sin C$$

Ex: $R = 5,\ A = 30^\circ \Rightarrow a = 2(5)\left(\frac{1}{2}\right) = 5.$

Grade - Low

Practice map, main concept: Sheet: C-II 3 | DPP: 2(4)

Also used in: none

Question 05

How to use sine law in proving identities?

Model Answer

Put $a = k\sin A,\ b = k\sin B,\ c = k\sin C$ (or $\sin A = \frac{a}{2R}$) to turn a side relation into an angle relation; then use $A + B + C = \pi$.

Ex: $\sin A\sin(B-C) = \sin(B+C)\sin(B-C) = \sin^2 B - \sin^2 C$, so $\sum a\sin(B-C) = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3, 9, 12; J-II 5-6; J-M 10; C-I 3; C-II 4, 14 | DPP: 1(1, 5, 8-11, 13-14, 16)

Also used in: Sheet: J-I 20, 29, 33; J-II 3, 30 | DPP: 1(7); 2(13)

Question 06

How to use sine law in triangle questions by dividing in two triangles?

Model Answer

A cevian AD splits the triangle; apply the sine rule in $\triangle ABD$ and $\triangle ADC$ and link them by the common side AD and $\sin\angle ADB = \sin\angle ADC$.

Ex: AD bisects A: $\dfrac{BD}{\sin\frac{A}{2}} = \dfrac{AD}{\sin B},\ \dfrac{DC}{\sin\frac{A}{2}} = \dfrac{AD}{\sin C} \Rightarrow \dfrac{BD}{DC} = \dfrac{\sin C}{\sin B} = \dfrac{c}{b}.$

Grade - High

Practice map, main concept: Sheet: J-I 10, 16; J-M 6; J-A 11; C-II 7 | DPP: 1(22)

Also used in: Sheet: J-I 35; C-II 2, 5, 28 | DPP: 1(21); 2(6)

Question 07

How to find ratio of sides when all angles are known?

Model Answer

$$a : b : c = \sin A : \sin B : \sin C$$

Ex: $A : B : C = 1 : 2 : 3 \Rightarrow 30^\circ, 60^\circ, 90^\circ \Rightarrow a : b : c = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 = 1 : \sqrt{3} : 2.$

Grade - Medium

Practice map, main concept: Sheet: J-I 13; J-II 28 | DPP: 1(19)

Also used in: Sheet: J-I 3, 15; J-M 9 | DPP: 2(7-8); 3(17)

Basic Laws : Cosine rule

Question 08

Write expression of cosine rule

Model Answer

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca}, \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Equivalently $a^2 = b^2 + c^2 - 2bc\cos A$ (and cyclic).

Grade - Medium

Practice map, main concept: Sheet: J-I 2, 6, 35; J-II 4; J-A 9 | DPP: 1(2, 7, 15); 3(18)

Also used in: DPP: 2(11)

Question 09

When should you use cosine rule?

Model Answer

(i) All three sides are known; (ii) two sides and the included angle are known; (iii) a relation in a^2, b^2, c^2 is given.

Ex: $a^2 + b^2 + ab = c^2 \Rightarrow \cos C = -\frac{1}{2} \Rightarrow C = 120^\circ$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 10

How to solve triangle if all three sides are known?

Model Answer

Find each angle by the cosine rule (largest angle first, it is opposite the largest side).

Ex: $a = 3, b = 5, c = 7$: $\cos C = \dfrac{9 + 25 - 49}{30} = -\dfrac{1}{2} \Rightarrow C = 120^\circ$.

Grade - High

Practice map, main concept: Sheet: J-M 4; C-I 5 | DPP: 1(17, 23); 2(5)

Also used in: Sheet: J-A 2-3

Question 11

How to find angles of triangle if ratio of all sides is known?

Model Answer

Take $a = pk,\ b = qk,\ c = rk$; k cancels in the cosine rule.

Ex: $a : b : c = 4 : 5 : 6 \Rightarrow \cos A = \frac{3}{4},\ \cos B = \frac{9}{16},\ \cos C = \frac{1}{8}$.

Grade - Low

Practice map, main concept: Sheet: J-I 4, 11; J-II 27; C-II 1

Also used in: none

Question 12

How to solve triangle if two sides and angle between them is known?

Model Answer

Third side by $a^2 = b^2 + c^2 - 2bc\cos A$; then the other angles by the sine rule (or tangent rule).

Ex: $b = 8, c = 5, A = 60^\circ \Rightarrow a^2 = 64 + 25 - 40 = 49 \Rightarrow a = 7$.

Grade - High

Practice map, main concept: Sheet: J-I 21; J-II 2, 36; J-A 6 | DPP: 2(15)

Also used in: Sheet: J-II 26; J-M 8; C-I 2; C-II 11

Question 13
How to determine nature of triangle if all three sides are known?

Model Answer
Let c be the largest side: $a^2+b^2>c^2$ acute, $=c^2$ right-angled, $<c^2$ obtuse-angled.
Ex: $4,5,6$: $16+25=41>36 \Rightarrow$ acute-angled.

Grade - High
Practice map, main concept: Sheet: J-I 8; J-M 9; J-A 5; C-I 17
Also used in: Sheet: J-I 1; J-II 20, 25; J-A 8

Question 14
What is the formula for $\cot A$ in terms of sides?

Model Answer

$$\cot A=\frac{b^2+c^2-a^2}{4\Delta}, \qquad \cot A+\cot B+\cot C=\frac{a^2+b^2+c^2}{4\Delta}$$

Ex: $a^2+b^2=101c^2 \Rightarrow \dfrac{\cot C}{\cot A+\cot B}=\dfrac{a^2+b^2-c^2}{2c^2}=50.$

Grade - Medium
Practice map, main concept: Sheet: J-II 1; J-A 10; C-II 10 | DPP: 1(24)
Also used in: Sheet: J-II 8; C-II 24

Basic Laws : Projection Formulae

Question 15
Write expression of projection formulae

Model Answer

$$a=b\cos C+c\cos B, \quad b=c\cos A+a\cos C, \quad c=a\cos B+b\cos A$$

Each side = sum of the projections of the other two sides on it.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 4 | DPP: 3(2)

Question 16
How to use projection formulae in questions?

Model Answer
Whenever $b\cos C+c\cos B$ type sums appear (also after writing $\cos^2\frac{C}{2}=\frac{1+\cos C}{2}$), replace them by the third side.
Ex: $a\cos^2\frac{C}{2}+c\cos^2\frac{A}{2}=\frac{a+c+(a\cos C+c\cos A)}{2}=\frac{a+b+c}{2}$; so $=\frac{3b}{2}\Rightarrow a,b,c$ in A.P.

Grade - High
Practice map, main concept: Sheet: J-I 20; J-II 3; J-M 3 | DPP: 1(3, 12)
Also used in: Sheet: J-A 9 | DPP: 1(11)

Basic Laws : Tangent Rule

Question 17

Write expression of tangent rule.

Model Answer

$$\tan\frac{B-C}{2}=\frac{b-c}{b+c}\cot\frac{A}{2},\quad \tan\frac{C-A}{2}=\frac{c-a}{c+a}\cot\frac{B}{2},\quad \tan\frac{A-B}{2}=\frac{a-b}{a+b}\cot\frac{C}{2}$$

Grade - Low

Practice map, main concept: Sheet: J-I 18-19; C-I 20; C-II 13 | DPP: 1(4)

Also used in: Sheet: J-I 12; C-II 31

Question 18

When should you use tangent rule?

Model Answer

When two sides and the included angle are known and the other angles (or $B-C$) are asked, without finding the third side.

Ex: $A=60^\circ,\ b:c=(\sqrt{3}+1):2\Rightarrow\tan\frac{B-C}{2}=\frac{\sqrt{3}-1}{\sqrt{3}+3}\cdot\sqrt{3}=2-\sqrt{3}\Rightarrow B-C=30^\circ.$

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 18

Area of Triangle

Question 19

Write expression for area of triangle.

Model Answer

$$\Delta=\tfrac{1}{2}bc\sin A=\tfrac{1}{2}ca\sin B=\tfrac{1}{2}ab\sin C=\frac{abc}{4R}=rs$$

Also $\Delta=\frac{1}{2}(\text{base})(\text{height})=\sqrt{s(s-a)(s-b)(s-c)}$.

Ex: $b=6,c=10,A=30^\circ\Rightarrow\Delta=\frac{1}{2}(6)(10)\frac{1}{2}=15.$

Grade - High

Practice map, main concept: Sheet: C-I 22

Also used in: Sheet: J-I 14, 22, 34; J-II 2, 21, 26, 33; J-M 2, 7-8; J-A 5-7, 14; C-I 13, 20; C-II 9, 13 | DPP: 3(3)

Question 20

How to find length of altitude if all sides are known?

Model Answer

Find Δ by Heron's formula; altitude on side a: $p_a=\dfrac{2\Delta}{a}$ (largest altitude on the smallest side).

Ex: $13,14,15$: $s=21,\ \Delta=\sqrt{21\cdot8\cdot7\cdot6}=84$; altitude on 14 is 12.

Grade - Low

Practice map, main concept: Sheet: J-I 5, 14; J-II 22, 32; C-I 15; C-II 23-25, 28 | DPP: 1(18); 3(3)

Also used in: Sheet: J-I 30

Half Angle formulae

Question 21

What is the formulae of $\sin\frac{A}{2}$, $\cos\frac{A}{2}$, $\tan\frac{A}{2}$ and area of triangle in terms of sides?

Model Answer

$$\sin\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{bc}},\quad \cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}}$$

$$\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}=\frac{\Delta}{s(s-a)},\quad \Delta=\sqrt{s(s-a)(s-b)(s-c)}$$

Ex: $13,14,15$, $a=13$: $\tan\frac{A}{2}=\frac{84}{21\cdot 8}=\frac{1}{2}$.

Grade - **High**

Practice map, main concept: Sheet: J-I 22-25; J-II 7-8; J-A 1, 8; C-I 12-13; C-II 21, 29-30 | DPP: 1(6, 20); 2(1, 11, 13-14); 3(1, 16-17)

Also used in: Sheet: J-I 5, 27; J-A 4; C-I 21-22; C-II 14, 19 | DPP: 3(12)

Question 22

How to derive formulae of $\sin\frac{A}{2}$, $\cos\frac{A}{2}$, $\tan\frac{A}{2}$ and area of triangle in terms of sides?

Model Answer

$2\sin^2\frac{A}{2}=1-\cos A=\frac{a^2-(b-c)^2}{2bc}=\frac{4(s-b)(s-c)}{2bc}$; similarly $2\cos^2\frac{A}{2}=1+\cos A=\frac{4s(s-a)}{2bc}$.

Then $\Delta=\frac{1}{2}bc\sin A=bc\sin\frac{A}{2}\cos\frac{A}{2}=\sqrt{s(s-a)(s-b)(s-c)}$.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Ambiguous Case of SOT

Question 23

Which case is known as ambiguous case?

Model Answer

When two sides and the angle opposite one of them are given (b,c,B): there may be no, one or two triangles.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 24

Derive third side in terms of two sides and the angle.

Model Answer

$$b^2=a^2+c^2-2ac\cos B\Rightarrow a^2-(2c\cos B)a+(c^2-b^2)=0$$

$$a=c\cos B\pm\sqrt{b^2-c^2\sin^2 B}$$

Two positive roots give two triangles; their difference is $2\sqrt{b^2-c^2\sin^2 B}$.

Grade - **Low**

Practice map, main concept: Sheet: J-II 10 | DPP: 2(2)

Also used in: Sheet: C-II 15

Question 25
Write all 4 conditions and no of triangles corresponding to each condition.

Model Answer
$b < c\sin B$: none; $b = c\sin B$: one (right angle at C); $c\sin B < b < c$: two; $b \geq c$: one.
Ex: $B = 30°,\ c = 8,\ b = 6$: $c\sin B = 4 < 6 < 8 \Rightarrow$ two triangles, $a = 4\sqrt{3} \pm 2\sqrt{5}$.

Grade - Low
Practice map, main concept: Sheet: J-II 9; C-I 1; C-II 15
Also used in: DPP: 2(2)

Question 26
Derive all 4 conditions with the help of triangle construction

Model Answer
Draw $BA = c$ and the ray BC at angle B; the distance of A from the ray is $c\sin B$. The circle (centre A, radius b) misses the ray, touches it, cuts it twice on the same side of B, or cuts it once (other point behind B).

Grade - Low
Practice map, main concept: none
Also used in: none

m-n theorem

Question 27
Write expression of m-n theorem

Model Answer
D on BC, $BD : DC = m : n$, $\angle ADC = \theta$, $\angle BAD = \alpha$, $\angle DAC = \beta$:

$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta = n\cot B - m\cot C$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 28
Derive expressions of m-n theorem

Model Answer
Sine rule in $\triangle ABD$ ($B = \theta - \alpha$) and $\triangle ADC$ ($C = \pi - \theta - \beta$): $\dfrac{m}{n} = \dfrac{\sin\alpha\,\sin(\theta+\beta)}{\sin(\theta-\alpha)\,\sin\beta}$.
Expand and divide by $\sin\alpha\sin\beta\sin\theta$: $m(\cot\alpha - \cot\theta) = n(\cot\beta + \cot\theta)$; the second form likewise with angles B, C.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 29
When would you use m-n theorem

Model Answer
When a line from a vertex divides the opposite side in a known ratio (median, trisection) and angles at A or at D are given or asked.

Ex: $B = 45°$, $BD : DC = 1 : 2$, $\angle BAD = 45° \Rightarrow \theta = 90°$: $0 = 2\cot 45° - \cot C \Rightarrow \cot C = 2$.

Grade - Low
Practice map, main concept: Sheet: C-II 5 | DPP: 1(21); 2(6)
Also used in: Sheet: C-II 7, 26

Circumcircle and Circumradius

Question 30
Define circumcircle, circumcenter, circumradius for a triangle?

Model Answer
Circumcircle: the circle through the three vertices; its centre O (circumcentre) is the meeting point of the perpendicular bisectors of the sides; its radius is R.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 31
Write expression to find circumradius in terms of side and sine of angle opposite to side

Model Answer

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

Ex: $c = 5,\ C = 30° \Rightarrow R = 5$.

Grade - High
Practice map, main concept: Sheet: J-I 29, 33; J-II 20, 30, 34; J-A 7, 14; C-I 2, 7, 10; C-II 9, 26 | DPP: 2(12); 3(11)
Also used in: Sheet: J-I 32, 36; J-II 17, 28; J-M 5, 7; J-A 6, 15; C-I 1, 15-16; C-II 8, 17, 23, 25, 29 | DPP: 2(4)

Question 32
Write expression to find circumradius in terms of sides and area of triangle

Model Answer

$$R = \frac{abc}{4\Delta}$$

Ex: $13, 14, 15$: $R = \dfrac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \dfrac{65}{8}$.

Grade - High
Practice map, main concept: none
Also used in: Sheet: J-II 29; J-A 3-4 | DPP: 3(4)

Question 33
What is the distance of circumcenter from all three vertices?

Model Answer
$OA = OB = OC = R$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 34
What is the distance of circumcenter from all three sides?

Model Answer
From BC: $R\cos A$; from CA: $R\cos B$; from AB: $R\cos C$ (a negative value means O is on the other side).
Ex: $OI \parallel BC \Rightarrow R\cos A = r \Rightarrow \cos B + \cos C = 1 + \frac{r}{R} - \cos A = 1$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 18 | DPP: 2(3)
Also used in: none

Question 35
Which point is at equal distance from all three vertices?

Model Answer
The circumcentre O (distance R).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 36
What is the position of circumcenter relative to a triangle?

Model Answer
Inside an acute-angled triangle, at the midpoint of the hypotenuse of a right-angled triangle, outside an obtuse-angled triangle.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 37
How to find position of circumcenter for a triangle geometrically?

Model Answer
Draw the perpendicular bisectors of any two sides; they meet at O.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 38
What is the angle subtended by a side at circumcenter?

Model Answer

$$\angle BOC = 2A, \quad \angle COA = 2B, \quad \angle AOB = 2C$$

So $\text{ar}(OBC) : \text{ar}(OCA) : \text{ar}(OAB) = \sin 2A : \sin 2B : \sin 2C$.

Grade - Low
Practice map, main concept: Sheet: J-II 37; C-I 16
Also used in: Sheet: C-II 27

Incircle and Inradius

Question 39
Define incircle, incenter, inradius for a triangle?

Model Answer
Incircle: the circle inside the triangle touching all three sides; its centre I (incentre) is the meeting point of the internal angle bisectors; its radius is r.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 40
Write all three expression to find inradius.

Model Answer

$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

Ex: $13, 14, 15$: $r = \frac{84}{21} = 4$.

Grade - High
Practice map, main concept: Sheet: J-I 30, 34; J-II 17, 24-26; J-M 7; J-A 3-4, 15; C-II 27 | DPP: 3(12)
Also used in: Sheet: J-I 26, 28; J-II 11, 13-14, 22-23, 27-29; J-A 6-7; C-I 10; C-II 6, 18-19, 25, 30 | DPP: 2(3, 10); 3(11, 16)

Question 41
What is the distance of incenter from all three vertices?

Model Answer

$$IA = r\,\text{cosec}\frac{A}{2}, \quad IB = r\,\text{cosec}\frac{B}{2}, \quad IC = r\,\text{cosec}\frac{C}{2}$$

Ex: equilateral, $r = \sqrt{3}$: $IA = \sqrt{3}\,\text{cosec}30^\circ = 2\sqrt{3}$.

Grade - Low
Practice map, main concept: Sheet: J-I 32; C-II 18 | DPP: 3(6)
Also used in: none

Question 42
What is the distance of incenter from all three sides?

Model Answer
Equal to r from each side.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 43
Which point is at equal distance from all three sides?

Model Answer
The incentre I (also each excentre, from the three side-lines).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 44
What is the position of incenter relative to a triangle?

Model Answer
Always inside the triangle.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 45
How to find position of incenter for a triangle geometrically?

Model Answer
Draw the internal bisectors of any two angles; they meet at I.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 46
What is the angle subtended by a side at incenter?

Model Answer

$$\angle BIC = \frac{\pi}{2} + \frac{A}{2}, \quad \angle CIA = \frac{\pi}{2} + \frac{B}{2}, \quad \angle AIB = \frac{\pi}{2} + \frac{C}{2}$$

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 37

Question 47

In what ratio, angle bisector of a vertex of triangle divides opposite side?

Model Answer

Internal bisector of A: $BD : DC = c : b$ (internally); external bisector divides BC externally in $c : b$.

Ex: $b = 4, c = 6, a = 5 \Rightarrow BD = \frac{6}{10} \cdot 5 = 3$.

Grade - **Low**

Practice map, main concept: Sheet: C-II 2

Also used in: Sheet: J-II 31; C-I 8

Question 48

In what ratio, incenter divides internal angle bisector?

Model Answer

$AI : ID = (b + c) : a$ (from the vertex).

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 49

Derive formula for inradius in terms of side and area of triangle.

Model Answer

$\Delta = \text{ar}(IBC) + \text{ar}(ICA) + \text{ar}(IAB) = \frac{1}{2}r(a + b + c) = rs \Rightarrow r = \dfrac{\Delta}{s}$.

Same splitting gives the radius of a circle centred on a side: centre on BC touching AB, AC: $\rho = \dfrac{2\Delta}{b + c}$.

Grade - **Low**

Practice map, main concept: Sheet: J-II 21; C-I 19

Also used in: none

Question 50

What is the position of incenter relative to a triangle?

Model Answer

Always inside the triangle (same as prompt 44).

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 51

How to find length of tangent from any vertex to incircle?

Model Answer

From A: $s - a$; from B: $s - b$; from C: $s - c$.

Ex: tangent lengths $4, 6, 8 \Rightarrow s = 18$, sides $a = 14, b = 12, c = 10$.

Grade - **High**

Practice map, main concept: Sheet: J-I 26; J-II 23, 29; J-M 5; J-A 2; C-II 19

Also used in: Sheet: J-II 24

Question 52
How to find position of incenter for a triangle geometrically?

Model Answer
Draw the internal bisectors of two angles; they meet at I (same as prompt 45).

Grade - Low
Practice map, main concept: none
Also used in: none

Excircle and exradii

Question 53
Define excircle, excenter, exradius for a triangle?

Model Answer
Excircle opposite A: touches BC and the extensions of AB, AC; centre I_1 (excentre) = meeting point of the internal bisector of A and the external bisectors of B, C; radius r_1.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 54
What is the position of excenter relative to a triangle?

Model Answer
Always outside the triangle; I_1 lies beyond side BC, opposite to A.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 55
How to find position of excenter for a triangle geometrically?

Model Answer
Draw the external bisectors of B and C; they meet at I_1 (on the internal bisector of A).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 56
Write all three expression to find exradius.

Model Answer

$$r_1 = \frac{\Delta}{s-a} = s\tan\frac{A}{2} = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

Similarly $r_2 = \frac{\Delta}{s-b}$, $r_3 = \frac{\Delta}{s-c}$. Ex: $13, 14, 15$: $r_1 = \frac{84}{8} = \frac{21}{2}$.

Grade - Low
Practice map, main concept: Sheet: J-I 27-28; J-II 11, 13-16, 19; C-II 17 | DPP: 2(10); 3(5, 10, 13, 15)
Also used in: Sheet: J-I 31; J-II 12, 22, 32; C-I 22 | DPP: 3(14)

Question 57
What is the distance of excenter from all three vertices?

Model Answer

$$I_1A = r_1 \operatorname{cosec}\frac{A}{2}, \quad I_1B = r_1 \sec\frac{B}{2}, \quad I_1C = r_1 \sec\frac{C}{2}$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 58
What is the distance of excenter from all three sides?

Model Answer

I_1 is at distance r_1 from BC and from the lines AB, AC.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 59
In what ratio, excenter divides internal angle bisector?

Model Answer

A, I, D, I_1 are collinear; I_1 divides AD externally in $(b + c) : a$ (I divides it internally in the same ratio).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 60
What is the angle subtended by a side at excenter?

Model Answer

$$\angle BI_1C = \frac{\pi}{2} - \frac{A}{2}$$

So $\angle BIC + \angle BI_1C = \pi$: B, I, C, I_1 are concyclic.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 61
What is the value of $\sum r_1$?

Model Answer

$$r_1 + r_2 + r_3 = 4R + r, \qquad r_1r_2 + r_2r_3 + r_3r_1 = s^2, \qquad rr_1r_2r_3 = \Delta^2$$

Also $\dfrac{1}{r_1} + \dfrac{1}{r_2} + \dfrac{1}{r_3} = \dfrac{1}{r}$. Ex: $r_1 = 8, r_2 = 12, r_3 = 24 \Rightarrow r = 4,\ s = \sqrt{576} = 24$.

Grade - Low

Practice map, main concept: Sheet: J-I 31 | DPP: 3(14)

Also used in: Sheet: J-II 15-16 | DPP: 3(15)

Question 62
When to use relation of r, r_1, r_2, r_3 in terms of R?

Model Answer
When R and angles appear together (identities like $r_1 - r = 4R\sin^2\frac{A}{2}$, $r_2 + r_3 = 4R\cos^2\frac{A}{2}$, $\sum r_1 - r = 4R$), or when the angles are known.

Ex: equilateral: $r = 4R\left(\frac{1}{2}\right)^3 = \frac{R}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 12, 18

Also used in: Sheet: J-II 19; J-A 4; C-I 17

Question 63
How to find length of tangent from any vertex to excircle?

Model Answer
Excircle opposite A: from A it is s; from B it is $s - c$; from C it is $s - b$.

Grade - Low

Practice map, main concept: none

Also used in: none

Centroid and Median

Question 64
Define centroid and median for a triangle?

Model Answer
Median: segment from a vertex to the midpoint of the opposite side; the three medians meet at the centroid G.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 65
In what ratio, a median divides the triangle in two parts?

Model Answer
Into two triangles of equal area $(1 : 1)$; the three medians make six equal parts.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 37

Question 66
What is the position of centroid relative to a triangle?

Model Answer
Always inside the triangle.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 67
In what ratio, centroid divides a median?

Model Answer
$AG : GD = 2 : 1$ (from the vertex).
Ex: median $AD = 4 \Rightarrow AG = \frac{8}{3}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 2
Also used in: Sheet: C-I 14 | DPP: 3(4)

Question 68
Write expression for length of median.

Model Answer

$$m_a = AD = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Ex: $3, 4, 5$: median to 5 is $\frac{1}{2}\sqrt{18 + 32 - 25} = \frac{5}{2}$.

Grade - Medium
Practice map, main concept: Sheet: C-I 4, 14; C-II 31 | DPP: 3(2, 4)
Also used in: Sheet: J-A 7

Length of Angle bisector

Question 69
Write expression for length of angle bisector.

Model Answer

$$AD = \frac{2bc}{b+c}\cos\frac{A}{2}$$

(From $\text{ar}(ABC) = \text{ar}(ABD) + \text{ar}(ADC)$.) Ex: $b = 3, c = 6, A = 120° \Rightarrow AD = \frac{36}{9} \cdot \frac{1}{2} = 2$.

Grade - Low
Practice map, main concept: Sheet: J-II 31; C-I 6; C-II 8
Also used in: Sheet: C-I 19

Orthocenter and Pedal Triangle

Question 70
Define altitude, orthocenter and pedal triangle for a triangle

Model Answer
Altitude: perpendicular from a vertex to the opposite side; altitudes meet at the orthocentre H; the triangle DEF of their feet is the pedal triangle.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 71
What is the distance of orthocenter from all three vertices?

Model Answer

$$HA = 2R\cos A, \quad HB = 2R\cos B, \quad HC = 2R\cos C$$

Ex: $AH = AO \Rightarrow 2R\cos A = R \Rightarrow A = 60°$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 36; C-I 11; C-II 20
Also used in: Sheet: J-II 36; C-I 9

Question 72
What is the distance of orthocenter from all three sides?

Model Answer

$$HD = 2R\cos B\cos C, \quad HE = 2R\cos C\cos A, \quad HF = 2R\cos A\cos B$$

Grade - **Low**
Practice map, main concept: Sheet: C-I 9; C-II 6
Also used in: Sheet: J-II 36; C-II 20

Question 73
What is the angle subtended by a side at orthocenter?

Model Answer

$$\angle BHC = \pi - A, \quad \angle CHA = \pi - B, \quad \angle AHB = \pi - C$$

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-II 37

Question 74
Write expression for sides and angles of pedal triangle.

Model Answer
Sides: $EF = a\cos A = R\sin 2A$, $FD = b\cos B$, $DE = c\cos C$; angles: $\pi - 2A$, $\pi - 2B$, $\pi - 2C$ (acute triangle).
H is the incentre of the pedal triangle; its area $= 2\Delta\cos A\cos B\cos C$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 35 | DPP: 3(7)
Also used in: none

Question 75
What is circumradius of a pedal triangle?

Model Answer
$\frac{R}{2}$ (it is the nine-point circle).
Ex: $R = 4 \Rightarrow$ circumradius of $DEF = 2$.

Grade - **Low**
Practice map, main concept: DPP: 2(9)
Also used in: none

Excentral Triangle

Question 76
Define excentral triangle for a triangle

Model Answer
The triangle $I_1I_2I_3$ formed by the three excentres.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 77
Relate excentral triangle with pedal triangle and how to find sides and angles of excentral triangle with its help

Model Answer
ABC is the pedal triangle of $I_1I_2I_3$ and I is its orthocentre (since $AI \perp I_2I_3$).
Angles $\frac{\pi}{2} - \frac{A}{2}$, $\frac{\pi}{2} - \frac{B}{2}$, $\frac{\pi}{2} - \frac{C}{2}$; circumradius $2R$; sides $I_2I_3 = 4R\cos\frac{A}{2}$ (cyclic).

Grade - Low
Practice map, main concept: none
Also used in: none

Regular Polygon

Question 78
How to find side and area of a regular polygon in terms of its circumradius

Model Answer

$$a = 2R\sin\frac{\pi}{n}, \qquad \text{Area} = \frac{n}{2}R^2\sin\frac{2\pi}{n}$$

Ex: hexagon: $a = 2R\sin 30^\circ = R$.

Grade - Medium
Practice map, main concept: Sheet: J-I 37-38; C-II 16, 22
Also used in: Sheet: J-I 39; J-M 1 | DPP: 1(23); 3(8-9)

Question 79
How to find side and area of a regular polygon in terms of its inradius

Model Answer

$$a = 2r\tan\frac{\pi}{n}, \qquad \text{Area} = nr^2\tan\frac{\pi}{n} = \frac{1}{4}na^2\cot\frac{\pi}{n}$$

Also $R + r = \frac{a}{2}\cot\frac{\pi}{2n}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 39; J-M 1 | DPP: 3(8-9)
Also used in: Sheet: J-I 38; C-II 16, 22

Question 80
How to find length of diagonal of a regular polygon in terms of its circumradius/inradius?

Model Answer

Diagonal joining vertices k steps apart subtends $\frac{2k\pi}{n}$ at the centre: $d_k = 2R\sin\dfrac{k\pi}{n}$, with $R = r\sec\dfrac{\pi}{n}$.

Ex: hexagon: shorter diagonal $2R\sin 60^\circ = \sqrt{3}R$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 81
What is internal angle of a regular polygon?

Model Answer

$$\frac{(n-2)\pi}{n}$$

Ex: hexagon: $\frac{4\pi}{6} = 120^\circ$.

Grade - Low

Practice map, main concept: none

Also used in: none

Important Points

Question 82
What is the relation between R and r for equilateral triangle?

Model Answer

$R = 2r$; and $r : R : r_1 = 1 : 2 : 3$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 83
Find all details of equilateral triangle in terms of its side.

Model Answer

$$h = \frac{\sqrt{3}}{2}a, \quad \Delta = \frac{\sqrt{3}}{4}a^2, \quad R = \frac{a}{\sqrt{3}}, \quad r = \frac{a}{2\sqrt{3}}, \quad r_1 = r_2 = r_3 = \frac{\sqrt{3}}{2}a$$

Ex: $r = 3 \Rightarrow a = 6\sqrt{3}$, perimeter $18\sqrt{3}$, $\Delta = 27\sqrt{3}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 84
Write expression for area of a cyclic quadrilateral

Model Answer

$$\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \qquad 2s = a+b+c+d$$

Ex: sides $2, 5, 3, 2$: $s = 6$, Area $= \sqrt{4 \cdot 1 \cdot 3 \cdot 4} = 4\sqrt{3}$.

Grade - **Low**

Practice map, main concept: Sheet: C-II 11

Also used in: none